

F-Actin Cytoskeleton Stain

Catalog #	Description
135-1305	F-Actin Blue 353/442 Cytoskeleton Stain
135-1306	F-Actin Violet 400/421 Cytoskeleton Stain
135-1307	F-Actin Green 493/517 Cytoskeleton Stain
135-1308	F-Actin Lime Green 520/547 Cytoskeleton Stain
135-1309	F-Actin Yellow 542/558 Cytoskeleton Stain
135-1310	F-Actin Orange 556/574 Cytoskeleton Stain
135-1311	F-Actin Red 590/618 Cytoskeleton Stain

For research purposes only.

Description

F-Actin Stain is an easy-to-use probe-based solution for visualizing filamentous actin structures in fixed mammalian cells by fluorescence microscopy.

F-actin is a major component of the cytoskeleton and is involved in fundamental cellular processes, such as cell division, morphogenesis, and migration. F-actin can be targeted by a well-recognized molecule called phalloidin. Phalloidin belongs to a family of toxins isolated from the fungus *Amanita phalloides*. It binds F-actin with high affinity at the interface between its subunits.

Bio-Rad's F-Actin Stain takes advantage of a probe, composed of a phalloidin molecule conjugated to a fluorophore, to help researchers detect the distribution of actin filaments in fixed cells (Figure 1).

F-Actin Cytoskeleton Stains are available in seven different excitation/emission wavelength combinations (Table 1) to enable multicolor experiments using the most common fluorescence cell imagers and microscopes.

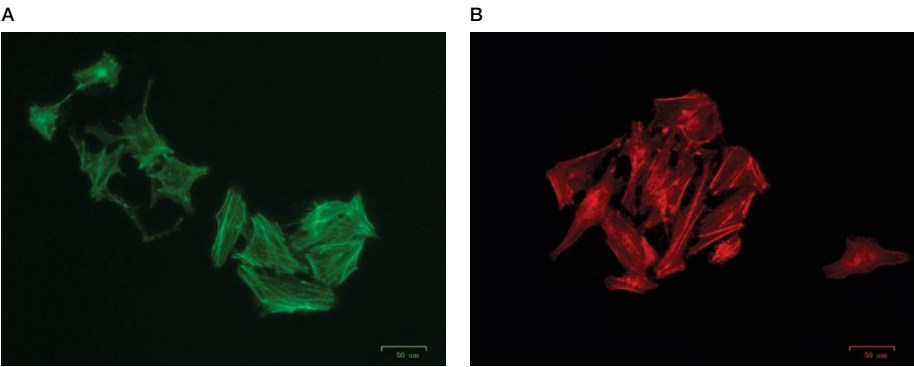


Fig. 1. Staining of filamentous actin with 1x F-Actin Green 493/517 (A) and 1x F-Actin Orange 556/574 (B). HeLa cells were fixed with formaldehyde, stained with F-Actin Stain, and visualized with the ZOE™ Fluorescent Cell Imager.

Contents and Storage

Each box contains three vials of F-Actin Cytoskeleton Stain (100 µl of 100x stain per vial). Store F-Actin Stain at –20°C. Protect from light and avoid repeated freeze-thaw cycles.

Ordering Information

Use Table 1 to select the appropriate F-Actin Stain for any given cell imaging experiment.

Table 1. Ordering information.

Catalog Number	Description	Optimal Channel on ZOE Fluorescent Cell Imager	Excitation, nm	Emission, nm
135-1305	F-Actin Blue 353/442	Blue	353	442
135-1306	F-Actin Violet 400/421	Blue	400	421
135-1307	F-Actin Green 493/517	Green	493	517
135-1308	F-Actin Lime Green 520/547	—	520	547
135-1309	F-Actin Yellow 542/558	Red	542	558
135-1310	F-Actin Orange 556/574	Red	556	574
135-1311	F-Actin Red 590/618	Red	590	618

Protocol

Preparing the Working Solution

1. Warm F-Actin Stain to room temperature and centrifuge briefly before opening.
2. Prepare a working solution of 1x F-Actin Stain by diluting the stock solution 1:100 in phosphate buffered saline (PBS) containing 1% bovine serum albumin.

Note 1: One vial (100 μ l) of F-Actin Stain is sufficient for making 10 ml of 1x working solution.

Note 2: Different cell types might stain differently. The concentration of working solution should be prepared accordingly.

Staining the Cells

1. Perform formaldehyde fixation. Incubate cells with 2–4% formaldehyde in PBS at room temperature for 20 min. The optimal incubation time may vary depending on cell type.
2. Rinse the fixed cells two to three times with PBS.
3. Optional: Add 0.1% Triton X-100 in PBS to fixed cells (after Step 2) for 3–5 min to increase permeability. Rinse the cells one to two times with PBS.
4. Add 1x F-Actin Stain (see Preparing the Working Solution section) to the fixed cells, and stain them at room temperature for 20–30 min. The optimal incubation time may vary depending on cell type.
5. Rinse the cells with PBS two to three times before mounting the sample.
6. Observe the cells using a ZOE Fluorescent Cell Imager or any fluorescence microscope equipped with an appropriate filter set and light source.

Visit bio-rad.com/web/F-ActinStain for more information.

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